Supplementary Material for

Certification without a safety dividend: forgeable credentials in populations of autonomous agents

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This document reports two results referred to by name in the main text, together with the extended parameter sweeps, the secondary policy experiments and the analytical derivations supporting it. All quantities are generated from the same version-controlled analysis pipeline, and every proof of a numbered statement in the main text appears in Section S7.

S1 Threshold sweeps

Supplementary Table S1 reports how dues and fines move the closed-form invasion and nucleation boundaries.

S2 No badge starts a club by itself

Everything above concerns a club that already exists. Whether one can form is a separate question with a blunt answer.

Proposition S1 (Badge futility)**.** *Against any resident whose conduct does not condition on badges, a certified design earns exactly its unbadged twin's payoff minus the dues, at every σ, every ρ and every L. Consequently, at $r = 0$ and $\kappa_g > 0$, no certified design invades any unconditional resident that its unbadged twin could not already invade.*

The reason is that a badge is a signal, and a signal changes nothing when nobody reads it [1]. A club must therefore bootstrap from encounters with itself, which is what assortment supplies.

Proposition S2 (Nucleation threshold)**.** *Against an unconditional unbadged resident (N, w, w) with $P(u,u) > P(v,w)$, a club (G, u, v) invades exactly when*

$$r \;>\; r^* \;=\; \frac{P(w,w) - P(v,w) + \kappa_g}{P(u,u) - P(v,w)} \;=\; 0.063 \tag{S1}$$

at the baseline, where the denominator is $P(\mathit{CS},\mathit{CS}) - P(\mathit{CAS},\mathit{CAS}) = 31.80$. The hypothesis is not decoration: the inequality is the sign of that denominator, and when it is negative assortment obstructs the club rather than enabling it. Against the safe *unbadged resident $(\mathsf{N}, \mathit{CS}, \mathit{CS})$ the denominator is -2.63 and the baseline club invades only* below $r = 0.240$. *When it vanishes the club invades at no assortment at all.*

The two residents are the two regimes of Observation 7, so the reversal is the statement that nucleating a club out of anarchy and nucleating one out of a settled safe uncertified market are different problems with opposite comparative statics in the one instrument available. Assortment is what lets a club form in an anarchic market, and it is also what lets a safe unbadged market resist one.

Table S1: Closed-form thresholds of the identity layer at zero liability. σ^* is the spoof success above which the unconditional forger invades the certified reciprocator club and σ_m the threshold for the behavioural mimic, both evaluated in a well-mixed population; r^* is itself an assortment, the one below which the baseline club cannot nucleate from the anarchic unconditional resident, whatever the verification. Note that σ_m exceeds σ^* in every row: well mixed, the exploiter invades first. The order reverses above $r = 0.077$ (Section 3.4), which is why the baseline $r = 0.1$ has $\sigma^* = 0.962 > \sigma_m = 0.938$ and why the manuscript reports the mimic as the first invader. Dues weaken the club; fines strengthen it only through the detections the screen already makes.

dues κ_g	fine ρ	σ^* (forger)	σ_m (mimic)	r^*
0.5	0	0.909	0.985	0.016
0.5	5	0.921	0.987	0.016
0.5	20	0.942	0.990	0.016
1.0	0	0.895	0.969	0.031
1.0	5	0.908	0.973	0.031
1.0	20	0.933	0.981	0.031
2.0	0	0.866	0.938	0.063
2.0	5	0.883	0.946	0.063
2.0	20	0.915	0.962	0.063
4.0	0	0.807	0.876	0.126
4.0	5	0.832	0.893	0.126
4.0	20	0.878	0.924	0.126
8.0	0	0.691	0.753	0.252
8.0	5	0.730	0.786	0.252
8.0	20	0.805	0.847	0.252

Taken together the two propositions say that verification quality and assortment are not substitutes. Attestation engineering, however good, moves σ, and σ does not appear in Equation (S1). What does appear is the dues, which is why a certification regime that is expensive to join can be unable to start even when it would be stable once running. Figure S1 shows the phase structure: panel A the switch at r^*, panel B the two-dimensional region in which a club can live, and panel C the asymmetry between keeping a club and building one.

That asymmetry is the sixth result. At $r = 0$ a club cannot re-enter a population that has collapsed either to forgery or to anarchy, at any σ whatsoever: the re-entry threshold does not exist. The conditions for holding a certification regime together are strictly weaker than the conditions for assembling one, so a regime that is allowed to collapse is not recovered by restoring the policy that sustained it.

S3 What the badge is doing, and what it is not

A mark that correlates with conduct could raise safety in two quite different ways. It could work as a *greenbeard*, by letting the bearers of a signal find one another; or it could work as a mere *assortment device*, by splitting encounters into two streams and letting the conduct game do the rest. At the baseline the two are indistinguishable, because the badge is both informative and correlated with conduct. Separating them needs a control that removes the information and keeps everything else, so we build one: each seat draws a badge independently of its conduct from a fixed distribution, which gives every design the same pass rate $\bar{q}$ and the same expected dues. The handshake, the four terms and their independence are the code path the badged model already uses; only the correlation is gone.

Table S2 scores four arms. Removing forgery ($\sigma = 0$) is the best case and removing the badge entirely is worse than the baseline, as expected. The informative comparison is the fourth

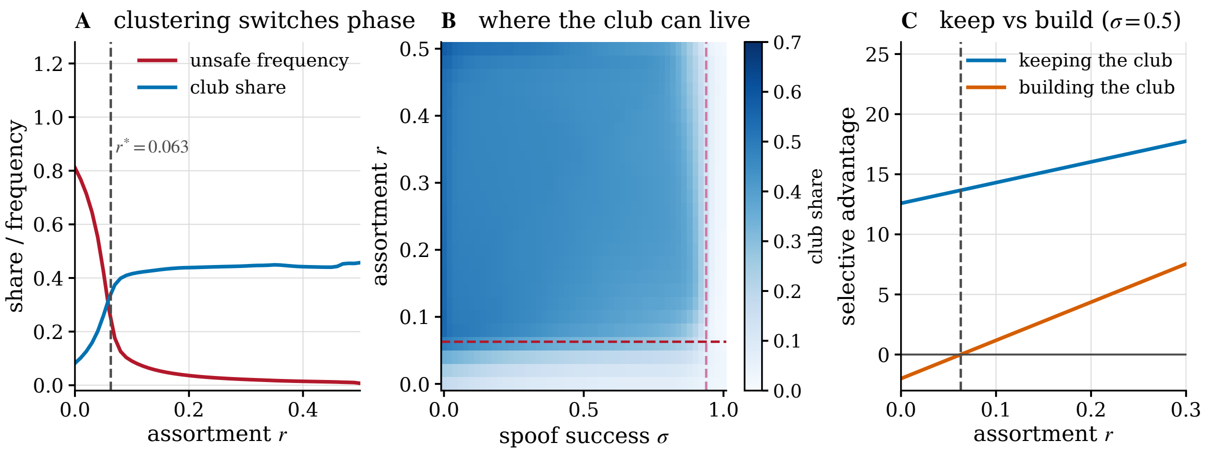

Figure S1: **Clubs need clustering, and losing one is worse than never having had it.** **(A)** Unsafe frequency and club share against assortment; the club switches on at $r^* = 0.063$ (Proposition S2). **(B)** Club share over spoof success and assortment. The horizontal line is r^* and the vertical line is the mimicry threshold σ_m; the club lives in the region between them. **(C)** The advantage of holding a club against forgery (blue) exceeds the advantage of building one from anarchy (orange) at every assortment, and the gap is what makes collapse irreversible.

Table S2: What the mark contributes, against three counterfactual marks. Entries are the stationary unsafe frequency of the small-mutation process at four assortments. The unforgeable arm is the best case and the no-badge arm the reference. The fourth arm keeps the badge, the dues and the handshake and removes only the correlation between the mark and the conduct behind it, so every design passes a check at the same rate; it is markedly worse than carrying no badge at all, because conduct still conditions on a check that has stopped discriminating.

design pool	$r = 0$	$r = 0.06$	$r = 0.1$	$r = 0.2$
full 48-design space	0.811	0.279	0.090	0.036
unforgeable badge ($\sigma = 0$)	0.746	0.226	0.067	0.032
no badge at all	1.000	0.667	0.159	0.050
badge drawn independently of conduct	0.988	0.619	0.488	0.191

arm, and it does not behave as a null model is supposed to.

A non-informative badge is not neutral. It is markedly *worse* than no badge at all: at $r = 0.1$ the unsafe frequency is 0.488 against 0.159 with no badge and 0.090 at the baseline. The reason is visible once stated. Agents still condition their conduct on the check, and the check has become a coin flip, so each seat now executes a conduct chosen at random rather than one chosen for the partner in front of it. Conditional reciprocity is precisely what that destroys. The damage is not the cost of the badge: an arm in which every badge is forged and free scores the same 0.488 as one that pays dues.

Two degenerate rates confirm that the construction is sound rather than broken, and they do so through Theorem 6. At $\bar{q} = 0$ no check ever passes, every seat executes s_{out}, and the arm reproduces the no-badge world exactly, at 0.158511 against 0.158511. At $\bar{q} = 1$ every check passes, every seat executes s_{in}, and the arm reproduces it again, to the same six digits, while paying full dues. The harm is therefore maximal at intermediate $\bar{q}$ and vanishes at both ends, which is the governance reading worth carrying away: a credential that everybody passes, or that nobody does, is merely useless, whereas a credential that passes half the time and is still acted upon is actively harmful.

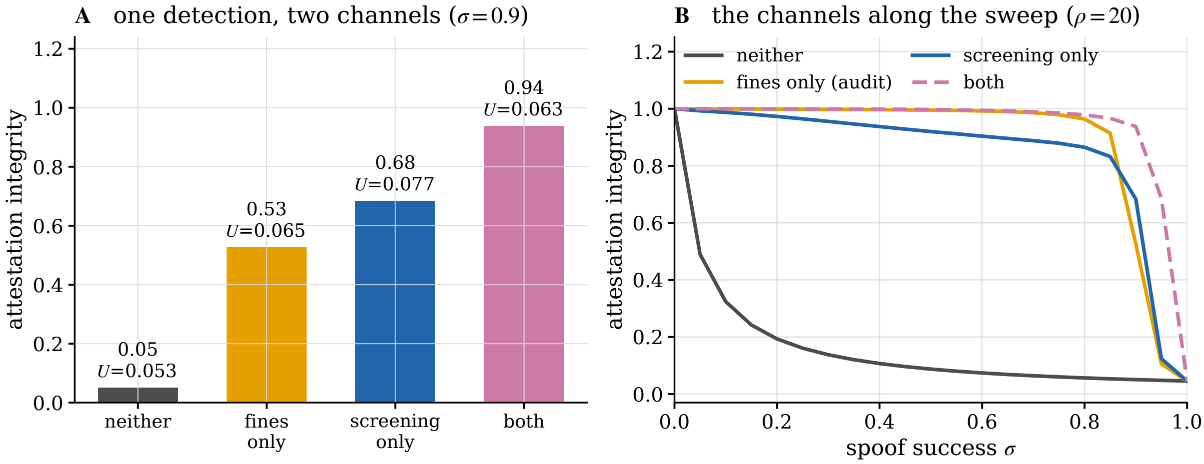

Figure S2: **The two consequences of one detection. (A)** Attestation integrity in the 2×2 at $\sigma = 0.9$, $\rho = 20$, with the unsafe frequency of each cell printed above the bar. Integrity varies by a factor of nearly twenty across cells in which the unsafe frequency barely moves. **(B)** The same four regimes along the spoof sweep.

S4 Screening and fining channels

One detection has two consequences, and governance regimes choose between them without always noticing. Real-time verification lets the counterparty act on a failed check; retrospective auditing levies a fine after the fact. The 2×2 of Figure S2 switches each off in turn at $\sigma = 0.9$ and $\rho = 20$.

The placebo cell, in which badges are checked but nothing follows from a failed check, leaves attestation integrity at 0.05: the population is essentially all forgers. Screening alone raises it to 0.68 and fines alone to 0.53; together they reach 0.94. The two channels are complements rather than substitutes, and neither reaches on its own what both reach together, which runs against the penalty-detection substitution of the standard enforcement model [2, 3].

The unsafe frequency tells a different story from the integrity, and this is the point of separating them. Across the four cells U moves only between 0.053 and 0.077, while integrity moves from 0.05 to 0.94. An evaluation regime that monitors conduct would rate all four cells as roughly equivalent and would be unable to distinguish a working attestation system from a completely hollow one. Attestation integrity has to be measured directly.

S5 Provider-scoped credentials

Current agent-identity approaches use incompatible credential formats, trust roots and delegation semantics [4]. We model one plausible consequence: marks issued per provider need not be recognised by another provider. Consider K symmetric provider clubs, each running the club design with its own mark, with cross-provider checks failing by construction.

Proposition S3 (Concentration frontier)**.** *With symmetric shares the within-provider encounter probability equals the Herfindahl index* HHI $= 1/K$*, and the ecosystem unsafe frequency is*

$$U \;=\; \mathrm{HHI}\, U(u,u) \;+\; (1-\mathrm{HHI})\, U(v,v) \;=\; 1-\mathrm{HHI} \tag{S2}$$

for the baseline club. Under federation, in which marks are mutually recognised [5], the ecosystem plays u throughout and $U = 0$ at any concentration.

A duopoly runs at $U = 0.500$ and eight providers at 0.875 (Table S3). Read naively this is an argument for concentration, which is not the reading we intend. It is an argument that

Table S3: The symmetric K-provider world under provider-scoped marks. Cross-provider checks fail, so the ecosystem unsafe frequency is $1 - \text{HHI}$ exactly; federated recognition plays the club design throughout and is safe at any concentration. The exclusion rent is the payoff gap between a member and a compliant unbadged entrant.

K	HHI	unsafe (scoped)	unsafe (federated)	member	rent
1	1.000	0.000	0.000	57.0	30.4
2	0.500	0.500	0.000	41.1	14.5
3	0.333	0.667	0.000	35.8	9.2
4	0.250	0.750	0.000	33.2	6.5
8	0.125	0.875	0.000	29.2	2.5
20	0.050	0.950	0.000	26.8	0.1

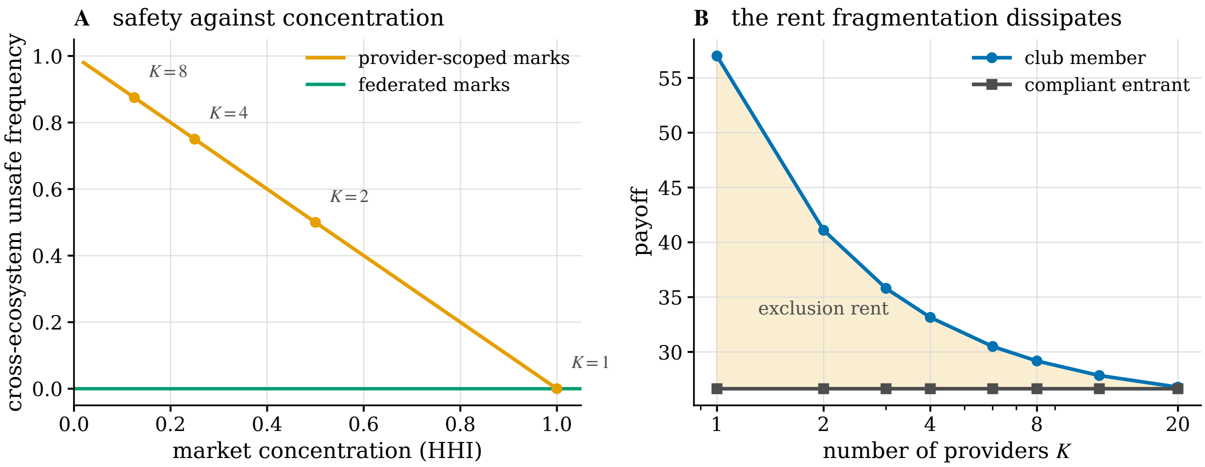

Figure S3: **Scoped marks tie safety to market structure.** **(A)** Cross-ecosystem unsafe frequency against concentration under provider-scoped marks is exactly $1 - \text{HHI}$; federated marks are safe at any concentration. **(B)** The exclusion rent, the payoff gap between a club member and a compliant unbadged entrant, dissipates as the market fragments, so the provider best placed to federate is the one least willing to.

scoping a safety mark to an issuer converts market structure into a safety variable, and that federation removes the dependence entirely.

The obstacle is that the incentive to federate runs backwards. Figure S3B shows the exclusion rent, the payoff gap between a club member and a compliant unbadged entrant that plays club conduct towards everyone and is treated as an outsider by every club. The rent is 30.4 under a single club and 2.5 across eight. A dominant provider therefore has the most to lose from mutual recognition [1], and a fragmented market has the least to gain from it while needing it most. The exclusion rent is also a competition concern in its own right: in the model, a compliant entrant is penalised 30.4 under a monopoly mark for no reason connected to its conduct. This is the kind of entry barrier that equitable-interoperability proposals seek to reduce [6].

S6 Liability and robustness

Proposition 2 predicts that identity should matter where liability cannot. Figure S4A measures it. The certification value, the unsafe frequency of the uncertified world minus that of the certified world at the same parameters, averages 0.054 below L_R and 0.0025 above it, a factor of 22. Only 15% of the (σ, L) grid shows a value above one percentage point, and that region

Table S4: The baseline equilibrium across nested design pools, in the small-mutation limit ($\sigma = 0.5$, $r = 0.1$, $\kappa_g = 2$, $L = 0$, $Z = 100$, $\beta = 0.05$). The identity layer roughly halves the long-run unsafe frequency of the uncertified world; forgery at a coin-flip spoof rate claws back about a quarter of the gain that unforgeable badges would deliver.

design pool	unsafe	club	falsebeard	integrity	social
certified world (48 designs)	0.090	0.42	0.07	0.92	39.1
unforgeable badges (32)	0.068	0.47	0.00	1.00	43.6
uncertified world (16)	0.159	0.00	0.00	N/A	25.4
raw race (4)	0.159	0.00	0.00	N/A	25.4

is concentrated at low liability and low spoof success.

The practical reading from our model is that attestation infrastructure is worth building where attribution is impossible. Agent-infrastructure work identifies traceability and accountability as central governance needs [7]; relevant settings include cross-border deployments, open-weight models with no traceable operator, and agent marketplaces with pseudonymous sellers. Where an operator can be identified and a liability regime applied [8], our model says the same money is better spent on that regime, which does not carry the exclusion cost of Section 3.5.

S6.1 Robustness

Table S5 and Figure S4 collect the checks.

Process. Across population sizes 50, 100, 200 and selection intensities 0.01, 0.05, 0.2, the baseline unsafe frequency ranges from 0.044 to 0.330. The upper end is the weak-selection corner ($\beta = 0.01$), where the process is close to neutral drift and no design is strongly favoured; the ordering of the results does not change.

Two dynamics. The small-mutation and replicator readings agree on the location and direction of the attestation collapse, with a correlation of 0.965 between their integrity curves along the σ sweep, the replicator figure being the basin average of the integrity at each attractor. They differ on the level of the unsafe frequency, and Section 3.5 explains why: they are answering different questions about a bistable flow, not disagreeing about one answer.

Payoff reading. Under the alternative reading of the setback clause, in which a realised setback removes only the prize and not the accumulated stage payoffs, every threshold moves but the phase structure does not: L_R rises from 0.551 to 1.486, σ^* falls from 0.866 to 0.676, σ_m is essentially unchanged at 0.934, and r^* rises from 0.063 to 0.095. Because the baseline $r = 0.1$ then sits just above r^*, the club is marginal at the baseline under this reading and the certified world runs at 0.933; raising assortment to $r = 0.3$ restores the club and yields a certification value of 0.598, the largest in the study. The honest summary is that the thresholds are what the model determines robustly, and that where a particular baseline sits relative to them is reading-dependent.

Closed forms. Every closed-form threshold agrees with brute-force root-finding on the exact matrices to 2×10^{-16}, across dues, fines and liabilities.

Design pools. Table S4 gives the baseline equilibrium across nested pools. Removing forgery from the design space lowers the baseline unsafe frequency from 0.090 to 0.068, so forgery at a coin-flip spoof rate costs about a quarter of what unforgeable badges would deliver. Removing badges entirely raises it to 0.159.

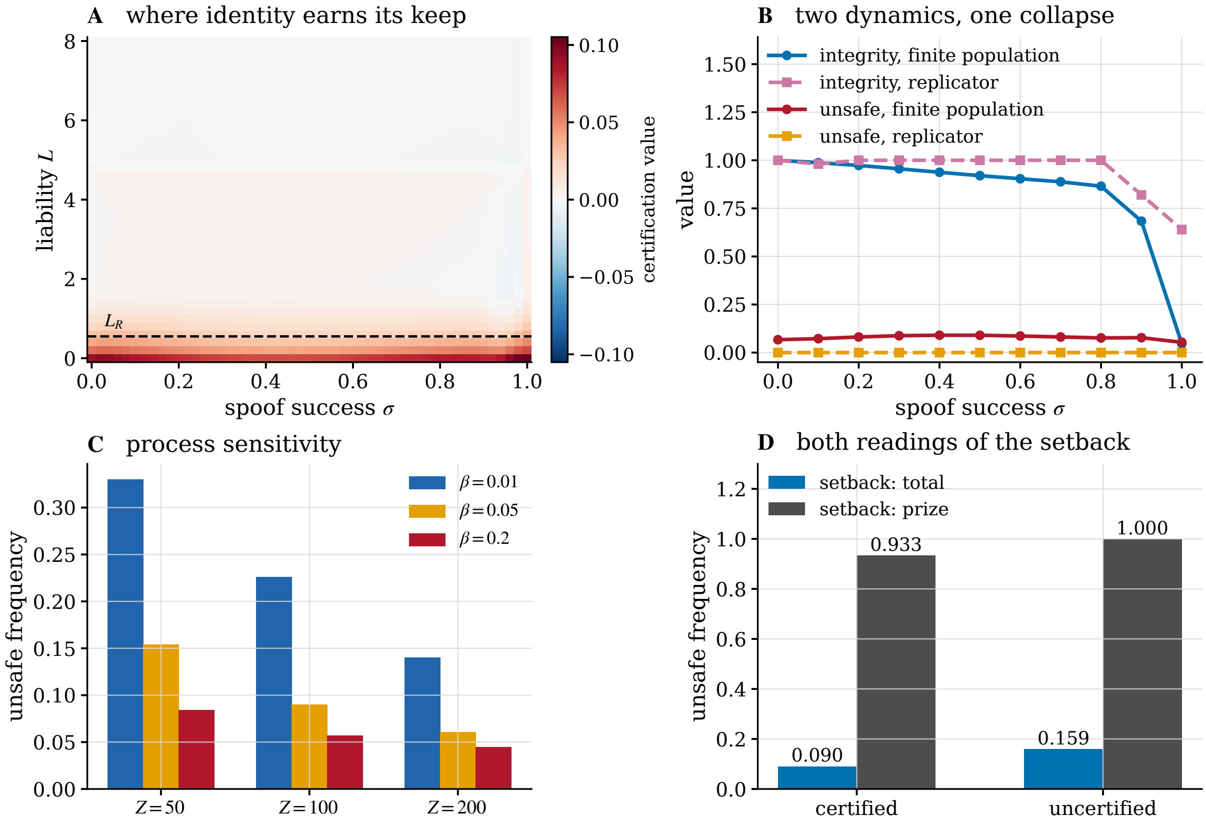

Figure S4: **Robustness.** **(A)** Certification value over spoof success and liability; the dashed line is L_R. The instrument earns its keep almost entirely below it. **(B)** The finite-population and replicator readings along the spoof sweep: they agree on the collapse of attestation integrity and disagree on the level of unsafe frequency, which is the finding of Section 3.5. **(C)** Population size and selection intensity. **(D)** Both readings of the setback clause.

S7 Analytical derivations

Proof of Proposition 2. Against a CS resident at $r = 0$, the private payoff of a rare CAS mutant is $P(\mathsf{CAS}, \mathsf{CS}) = A(\mathsf{CAS}, \mathsf{CS}) - L\,M(\mathsf{CAS}, \mathsf{CS})$ and the resident earns $P(\mathsf{CS}, \mathsf{CS})$. The difference is affine in L with slope $-[M(\mathsf{CAS}, \mathsf{CS}) - M(\mathsf{CS}, \mathsf{CS})] < 0$, so it is positive exactly below the ratio in Equation (8). With assortment, the exact difference against the canonical safe resident is the joint expression in Equation (9); the two intercepts cannot be combined into a rectangle. For the general statement, a safe unbadged resident plays a safe design in self-play; since badges are absent, every check fails and only s_{out} is ever executed, so the resident is behaviourally one of $(\mathsf{N}, \cdot, \mathsf{AS})$ or $(\mathsf{N}, \cdot, \mathsf{CS})$. With assortment r, the mutant meets itself with probability r, so the invasion calculation above gives its advantage over a $(\mathsf{N}, \cdot, \mathsf{CS})$ resident at $L = 0$ as $r\,A(\mathsf{CAS}, \mathsf{CAS}) + (1-r)\,A(\mathsf{CAS}, \mathsf{CS}) - A(\mathsf{CS}, \mathsf{CS})$, which is affine and decreasing in r and vanishes at Equation (10). The same calculation against a $(\mathsf{N}, \cdot, \mathsf{AS})$ resident gives $42.77 - 74.57\,r$, which vanishes only at $r = 0.574$, so the $(\mathsf{N}, \cdot, \mathsf{CS})$ residents are the binding ones and $r^{\dagger}$ is the smaller of the two roots. The invasion of $(\mathsf{N}, \mathsf{CAS}, \mathsf{CAS})$ against each of the eight such residents is verified directly on the exact matrices at $r = 0$, and the four $(\mathsf{N}, \cdot, \mathsf{CS})$ residents are verified to resist at the baseline $r = 0.1$. □

Proof of Proposition 3. In a monomorphic club, a rare unconditional forger (F, w, w) meets club members. Its badge passes with probability σ, in which case the member plays u; otherwise the member plays v. The forger plays w regardless. Its payoff is therefore $\sigma P(w, u) + (1 - \sigma)P(w, v) - \kappa_f - (1 - \sigma)\rho$ and the resident earns $P(u, u) - \kappa_g$. Both are affine in σ; equating

Table S5: Process robustness of the baseline equilibrium. Upper block: population size and selection intensity. Lower block: both readings of the setback clause of the race, where 'total' means a realised setback removes the accumulated stage payoffs as well as the prize and 'prize' means it removes only the prize. The two blocks do not share the meaning of the last column, so each carries its own header row.

Z	β	unsafe	club	integrity
50	0.01	0.330	0.28	0.78
50	0.05	0.154	0.39	0.88
50	0.2	0.084	0.42	0.92
100	0.01	0.226	0.35	0.84
100	0.05	0.090	0.42	0.92
100	0.2	0.057	0.45	0.94
200	0.01	0.140	0.39	0.89
200	0.05	0.060	0.43	0.94
200	0.2	0.044	0.46	0.95
setback scope		certified unsafe	club share	uncertified unsafe
total		0.090	0.42	0.159
prize		0.933	0.03	1.000

and solving gives Equation (11) whenever $D = P(w,u) - P(w,v) + \rho \neq 0$. This D is the slope of the forger's advantage: for $D > 0$ that advantage increases with σ and the club resists below the threshold; for $D < 0$ it decreases and the inequality reverses. Finally, $D = 0$ exactly when $\rho = P(w,v) - P(w,u)$, in which case the advantage is independent of σ and no division is made. □

Proof of Proposition 4. Solve the same indifference condition for ρ instead of σ. The numerator of Equation (12) is the forger's advantage at zero fine, and the denominator $1-\sigma$ is the probability that a fine is levied. If the numerator is positive at $\sigma = 1$, that is if $P(w,u) - \kappa_f > P(u,u) - \kappa_g$, then $\rho^* \to \infty$. □

Proof of Proposition S1. Let the resident be (N, w, w). Its conduct does not depend on the focal badge, so it plays w whatever the focal design carries. The focal design's own check of the resident always fails, since the resident carries no badge, so the focal plays s_{out}. The encounter is therefore (s_{out}, w) for both a certified and an unbadged focal design with the same conduct pair, and the payoffs differ only by κ_g. Since σ enters only through the pass rate of a forged badge, which is not present, the identity holds at every σ; the same argument gives independence of ρ and L enters both sides identically. The conclusion follows because at $r = 0$ a rare mutant meets only residents. □

Proof of Proposition S2. With assortment r, a rare club mutant meets itself with probability r and the resident otherwise. Its fitness is $r[P(u,u) - \kappa_g] + (1-r)[P(v,w) - \kappa_g]$, since in a self-encounter both badges pass and both play u, while against the unbadged resident the club plays v and the resident plays w. The resident earns $P(w,w)$. Rearranging the inequality gives Equation (S1). □

Proof of Proposition 5. The mimic (F, u, v) in a club population always sees the genuine resident badge pass and therefore plays u. The resident plays u towards the mimic when its forged badge passes and v when it does not. Its expected payoff is $\sigma P(u,u) + (1-\sigma)P(u,v) - \kappa_f - (1-\sigma)\rho$ against the resident's $P(u,u) - \kappa_g$. Solving for the indifference point gives Equation (14). The mimic's threshold lies below the exploiter's whenever the exploiter's conduct is worse in self-play than the club's, which is what makes the exploiter pay a larger assortment penalty. □

Proof of Proposition S3. Under symmetric shares, an agent of provider k meets an agent of the same provider with probability $1/K$, which equals $\mathrm{HHI} = \sum_{k=1}^{K}(1/K)^2 = 1/K$. Within-provider checks pass and both play u; cross-provider checks fail and both play v. Equation (S2) is the resulting mixture. For the baseline club, $U(\mathsf{CS}, \mathsf{CS}) = 0$ and $U(\mathsf{CAS}, \mathsf{CAS}) = 1$, giving $U = 1 - \mathrm{HHI}$. Under federation every check passes and the mixture collapses to $U(u, u)$. □

Proof of Proposition 9. Take $r = 0$, so that a rare entrant meets only members. A member of a monomorphic club meets a member at every encounter, both checks pass and both execute u, so its private payoff is $P(u, u) - \kappa_g$. The entrant carries no badge, so every member's check of it fails and every member executes v, while the entrant plays u towards everybody by construction and therefore earns $P(u, v)$ with no dues. The entry penalty is the difference, $P(u, u) - \kappa_g - P(u, v)$, and the boundary unsafe frequency is $U(u, v)$, neither of which involves σ. With $u = \mathsf{CS}$ and $v = \mathsf{CAS}$ this is $59.00 - 2.00 - 26.64 = 30.36$ and $U(\mathsf{CS}, \mathsf{CAS}) = 0.461$; with $v = \mathsf{CS}$ the encounter is $(\mathsf{CS}, \mathsf{CS})$, so the penalty is $59.00 - 2.00 - 59.00 = -2.00$ and the boundary is safe. At the baseline assortment $r = 0.1$ the entrant's own self-encounters lift its payoff and the penalty falls to 27.12; the sign and the ordering across out-group policies are unchanged. □

Proof of Theorem 6. Both faces are invariant because Equation (5) gives $\dot{x}_i = 0$ wherever $x_i = 0$, so a coordinate that starts at zero stays there and the support of a state can only shrink.

Fix a state supported on Δ_{G}. Every design in the support carries a genuine badge, so $q_j = 1$ for every such j, and the handshake weights of Equation (1) are $w_{\mathrm{in}}(q_j) = 1$ and $w_{\mathrm{out}}(q_j) = 0$. Three of the four terms of the sum therefore vanish and $a(i, j) = A(s^i_{\mathrm{in}}, s^j_{\mathrm{in}})$. The same collapse applies to m and u with M and U in place of A, since those matrices are the same sum with a different kernel. No design in the support carries a forged badge, so the expected fine is zero and every design pays κ_g. Substituting into Equation (2) gives the first half of Equation (15). On Δ_{N} every check fails, $q_j = 0$, the surviving term is $(\mathrm{out}, \mathrm{out})$, and no badge is bought, which gives the second half.

For the reduction, let $y_c = \sum_{i: s^i_{\mathrm{in}} = c} x_i$ be the conduct marginal on Δ_{G}. By Equation (15) the payoff of design i depends on i only through s^i_{in}, so Equation (4) gives a common value $g_c = r\,[P(c, c) - \kappa_g] + (1 - r)\sum_d y_d\,[P(c, d) - \kappa_g]$ for every design with $s^i_{\mathrm{in}} = c$, and $\bar{f} = \sum_d y_d g_d$. Summing Equation (5) over the designs sharing a conduct yields $\dot{y}_c = y_c(g_c - \bar{f})$, a four-strategy replicator equation in its own right. The same computation on Δ_{N} with s_{out} gives the identical system without the κ_g terms. Finally, adding a constant γ to every entry of the payoff matrix sends $g_c \mapsto g_c + \gamma$ and $\bar{f} \mapsto \bar{f} + \gamma$, because $r + (1 - r)\sum_d y_d = 1$, so it leaves $g_c - \bar{f}$ and hence the flow unchanged. The two marginal systems are therefore the same flow, and rest points, stability and attractors coincide. At corresponding rest points the realised conduct profile is the same, so U is the same; the social functional of Equation (3) differs only by the badge cost, which is κ_g on one face and 0 on the other. □

Proof of Lemma 1. Write the handshake of Equation (1) as a sum over the focal's executed branch x and the partner's executed branch y. Averaging against a population state gives

$$\sum_j x_j\, a(i, j) = \sum_{x,y} w_y(q_i) \sum_c \Big[\sum_j x_j\, w_x(q_j)\, \mathbf{1}[s^j_y = c]\Big] A(s^i_x, c).$$

The bracket is $Q_{y,c}(x)$ when $x = \mathrm{in}$ and $T_{y,c}(x) - Q_{y,c}(x)$ when $x = \mathrm{out}$, so the whole expression is a linear combination of the sixteen functionals of Equation (7) with coefficients depending only on i. The liability, badge-cost and fine terms of Equation (2) are constants in j and are absorbed by $\sum_c T_{y,c} = 1$. Hence every row of π_P lies in the span. Three independent linear relations hold on that span: $\sum_c T_{\mathrm{in},c} = 1$, $\sum_c T_{\mathrm{out},c} = 1$, and $\sum_c Q_{\mathrm{in},c} = \sum_c Q_{\mathrm{out},c} = \sum_j x_j q_j$, since each side counts the mean pass rate once. Sixteen functionals subject to three relations leave thirteen free directions, and the rank is exactly thirteen rather than at most thirteen because the four conducts of the race are pairwise distinct in A, so no further relation holds. A least-squares fit of π_P onto the sixteen rows leaves a residual below 1.9×10^{-13} at every sampled (σ, r), and rank $\pi_P = 13$ numerically at each of them. $\square$

Proof of the harm decomposition of Section 2.4. The decomposition is an identity, not an approximation. Let w_{ij} be the probability of the ordered pair (i, j) under the encounter law and let $\mathcal{B}$ partition the ordered pairs by the badge status of the two seats. Then $U = \sum_{ij} w_{ij} u(i, j) = \sum_{B \in \mathcal{B}} \sum_{(i,j) \in B} w_{ij} u(i, j)$, so the block contributions sum to U exactly. The zero entries are exact in the model: a club member facing a club member executes $(\mathsf{CS}, \mathsf{CS})$, which contains no unsafe action, and the surviving unbadged designs execute $(\mathsf{CS}, \mathsf{CS})$ against each other for the same reason. $\square$

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